



THE DIGITAL MATH FRAMEWORK AS A ROBUST AND GENERAL SCIENTIFIC BASIS FOR SOCIETY - ADVANCEMENTS IN THE ADAPTIVE EULER BREAKTHROUGH FOR AERODYNAMICS AND DIGIMAT EDUCATIONAL PROGRAM

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Today we see large-scale threats to the scientific basis of modern society, which often contain elements of attacks on the concept of truth. We develop the Digital Math framework - where FEniCS is a key component - to form a robust general scientific basis for research and education, and a counter to the threats to science by guaranteeing reproducibility and falsifiability.

All of natural science and engineering is formulated as mathematical equations - models, and the Digital Math framework provides powerful and at the same time simple and reproducible methods and tools for predicting/solving the models, and pedagogical frameworks for learning the methods and models. Key components are the FEniCS framework for automated computational modeling and the DigiMat motivational educational framework.

We here contribute advancements in the Adaptive Euler methodology for first-principles reliable and efficient aerodynamics - solving the Grand Challenge of aerodynamics, and the Unified Continuum methodology for general and robust continuum modeling such as fluid-structure interaction (FSI).

We present specific developments of Adaptive Euler and DigiMat:

1. A simple and general Adaptive Euler formulation validated in the High Lift, Drag and Aeroelasticity Prediction workshops, including compressible phenomena such as shocks, resolving the Grand Challenge of aerodynamics with fast and cheap first-principles aerodynamics simulation.
2. Development toward a general and first-principles Adaptive Euler Unified Continuum FSI formulation with the ability to predict aeroelastic phenomena such as: flutter and buffeting (wake-tail and transonic).
3. Industrial workflows and open model results developed together with SAAB and FOI.
4. DigiMat pedagogical learning activities developed in collaboration with the Technical Museum and IVA Royal Academy of Engineering Sciences Technology Leap.

For references and more details, see: <http://digitalmath.tech/fenics2026-JohanJansson/>

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